BITS PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI
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Data Structures & Algorithms (CS F211)

```c
/*
Name: Kirtika
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Course: CS F211 - Data Structures & Algorithms
Lab Exercise 7: Quick Sort on Student Records
*/

#include <stdio.h>
#include <string.h>
#include <sys/time.h>

#define MAX_STUDENTS 200
#define NAME_LEN 64

typedef struct {
    char name[NAME_LEN];
    int id;
    float cgpa;
} Student;

static long long current_time_us(void) {
    struct timeval tv;
    gettimeofday(&tv, NULL);
    return (long long)tv.tv_sec * 1000000LL + (long long)tv.tv_usec;
}

static void print_records(const Student arr[], int n) {
    printf("%-5s %-20s %-10s %-10s\n", "No.", "Name", "ID", "CGPA");
    printf("-----------------------------------------------\n");
    for (int i = 0; i < n; i++) {
        printf("%-5d %-20s %-10d %-10.2f\n", i + 1, arr[i].name, arr[i].id,
arr[i].cgpa);
    }
}

static void print_partition_stage(const Student arr[], int n, int low, int high,
float pivot, int pivot_index) {
    printf("\nAfter partition [%d..%d], pivot = %.2f at index %d\n", low, high,
pivot, pivot_index);
    for (int i = 0; i < n; i++) {
        printf("%s(%.2f)", arr[i].name, arr[i].cgpa);
        if (i != n - 1) {
            printf(" ");
        }
    }
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    printf("\n");
}

static void swap_students(Student *a, Student *b) {
    Student temp = *a;
    *a = *b;
    *b = temp;
}

static int partition(Student arr[], int low, int high, int total_size) {
    float pivot = arr[high].cgpa;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (arr[j].cgpa <= pivot) {
            i++;
            swap_students(&arr[i], &arr[j]);
        }
    }

    swap_students(&arr[i + 1], &arr[high]);
    int pivot_index = i + 1;

    print_partition_stage(arr, total_size, low, high, pivot, pivot_index);

    return pivot_index;
}

static void quick_sort(Student arr[], int low, int high, int total_size) {
    if (low < high) {
        int pi = partition(arr, low, high, total_size);
        quick_sort(arr, low, pi - 1, total_size);
        quick_sort(arr, pi + 1, high, total_size);
    }
}

int main(void) {
    Student students[MAX_STUDENTS];
    int n;

    printf("Enter number of student records (at least 10, max %d): ",
MAX_STUDENTS);
    if (scanf("%d", &n) != 1 || n < 10 || n > MAX_STUDENTS) {
        printf("Invalid number of records.\n");
        return 1;
    }

    printf("Enter each record as: name id cgpa\n");
    for (int i = 0; i < n; i++) {
        printf("Record %d: ", i + 1);
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        if (scanf("%63s %d %f", students[i].name, &students[i].id,
&students[i].cgpa) != 3) {
            printf("Invalid input format.\n");
            return 1;
        }
    }

    printf("\nUnsorted student records:\n");
    print_records(students, n);

    long long start = current_time_us();
    quick_sort(students, 0, n - 1, n);
    long long elapsed = current_time_us() - start;

    printf("\nSorted student records (ascending by CGPA):\n");
    print_records(students, n);

    printf("\nQuick sort execution time: %lld micro-secs\n", elapsed);
    return 0;
}
```

OUTPUT

```
manavaryasingh@MacBook-Pro-5 kritika-dsa % cd "/Users/manavaryasingh/kritika-dsa"
gcc "week 7/ex7_quick_sort_students.c" -o "week 7/ex7"
"week 7/ex7"
Enter number of student records (at least 10, max 200): 10
Ria 2001 3.12
Karan 2002 2.95
Neha 2003 3.80
Amit 2004 3.45
Zeeshan 2005 3.01
Bina 2006 3.99
Charu 2007 2.50
Deep 2008 3.45
Esha 2009 3.45
Faisal 2010 2.75Enter each record as: name id cgpa
Record 1: Record 2: Record 3: Record 4: Record 5: Record 6: Record 7: Record 8: Record 9: Record 10:

Unsorted student records:
No.   Name                 ID         CGPA
-----------------------------------------------
1     Ria                  2001       3.12
2     Karan                2002       2.95
3     Neha                 2003       3.80
4     Amit                 2004       3.45
5     Zeeshan              2005       3.01
6     Bina                 2006       3.99
7     Charu                2007       2.50
8     Deep                 2008       3.45
9     Esha                 2009       3.45
10    Faisal               2010       2.75

After partition [0..9], pivot = 2.75 at index 1
Charu(2.50) Faisal(2.75) Neha(3.80) Amit(3.45) Zeeshan(3.01) Bina(3.99) Ria(3.12) Deep(3.45) Esha(3.45) Karan(2.95)

After partition [2..9], pivot = 2.95 at index 2
Charu(2.50) Faisal(2.75) Karan(2.95) Amit(3.45) Zeeshan(3.01) Bina(3.99) Ria(3.12) Deep(3.45) Esha(3.45) Neha(3.80)

After partition [3..9], pivot = 3.80 at index 8
Charu(2.50) Faisal(2.75) Karan(2.95) Amit(3.45) Zeeshan(3.01) Ria(3.12) Deep(3.45) Esha(3.45) Neha(3.80) Bina(3.99)

After partition [3..7], pivot = 3.45 at index 7
Charu(2.50) Faisal(2.75) Karan(2.95) Amit(3.45) Zeeshan(3.01) Ria(3.12) Deep(3.45) Esha(3.45) Neha(3.80) Bina(3.99)

After partition [3..6], pivot = 3.45 at index 6
Charu(2.50) Faisal(2.75) Karan(2.95) Amit(3.45) Zeeshan(3.01) Ria(3.12) Deep(3.45) Esha(3.45) Neha(3.80) Bina(3.99)

After partition [3..5], pivot = 3.12 at index 4
Charu(2.50) Faisal(2.75) Karan(2.95) Zeeshan(3.01) Ria(3.12) Amit(3.45) Deep(3.45) Esha(3.45) Neha(3.80) Bina(3.99)

Sorted student records (ascending by CGPA):
No.   Name                 ID         CGPA
-----------------------------------------------
1     Charu                2007       2.50
2     Faisal               2010       2.75
3     Karan                2002       2.95
4     Zeeshan              2005       3.01
5     Ria                  2001       3.12
6     Amit                 2004       3.45
7     Deep                 2008       3.45
8     Esha                 2009       3.45
9     Neha                 2003       3.80
10    Bina                 2006       3.99

Quick sort execution time: 67 micro-secs
```